Arista Access Node MAR-ACC-36 Standard Operating Procedure for ALARM_4304: Major Routing Protocol Flap due to BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_4304

Description

This SOP outlines the procedures to diagnose and resolve a major routing protocol flap failure on Arista Access Node MAR-ACC-36 located at the Marathahalli Core site. The failure is characterized by a routing protocol flap, which can be attributed to BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. This failure domain occurs due to the inherent complexities of routing protocols, specifically the dynamic nature of BGP, OSPF, and ISIS, which can lead to flapping when faced with network changes, misconfigurations, or hardware issues. The Access Node, being a critical component of the network architecture, is expected to operate within predetermined boundaries of routing protocol stability. The occurrence of a major routing protocol flap on this node necessitates immediate attention to prevent network instability and potential service disruptions.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible to dispatch remote hands for on-site assistance, if required.

- Check the status of redundant links to ensure that there are no issues with link connectivity or redundancy.

- Confirm that all necessary network administrators are aware of the failure and are available to assist in the resolution process.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP table version is 123, main routing table version is 123 BGP namespace is 123 Local AS number on this router is 123 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State | 192.168.1.1 4 123 123 123 123 0 0 00:00:10 Established |, Description: This command displays a summary of the BGP routing
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table, including the BGP table version, main routing table version, and local AS number. It also shows the number of messages received and sent, as well as the table version, input and output queues, and up/down time.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface | 192.168.1.1 1 Full/ - 00:00:32 192.168.1.1 Ethernet1/1 |, Description: This command displays a list of OSPF neighbors, including their ID, priority, state, dead time, address, and interface.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Sequence Checksum Length | 123.456.789.0 1234567890 1234567890 123 | ISIS SPF Database: SPF ID Sequence Checksum Length | 123.456.789.0 1234567890 1234567890 123 |, Description: This command displays the ISIS database, including the LSP ID, sequence number, checksum, and length. It also shows the SPF ID, sequence number, checksum, and length.
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Resolution Steps

Step 1: Step: Clear BGP session, Command: clear bgp *, Description: Clear the BGP session to resolve BGP route churn issues. This command will reset the BGP session and re-establish new connections.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf adjacency *, Description: Reset the OSPF adjacency to resolve OSPF adjacency loss issues. This command will reset the OSPF adjacency and re-establish new connections.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear the ISIS LSP database to resolve ISIS LSP flooding issues. This command will reset the ISIS LSP database and re-establish new connections.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version is 123, main routing table version is 123 BGP namespace is 123 Local AS number on this router is 123 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State | 192.168.1.1 4 123 123 123 123 0 0 00:00:10 Established |, Description: Verify that the BGP session is established and the routing table is stable.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface | 192.168.1.1 1 Full/ - 00:00:32 192.168.1.1 Ethernet1/1 |, Description: Verify that the OSPF adjacency is established and the neighbor is reachable.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Sequence Checksum Length | 123.456.789.0 1234567890 1234567890 123 | ISIS SPF Database: SPF ID Sequence Checksum Length | 123.456.789.0 1234567890 1234567890 123 |, Description: Verify that the ISIS LSP database is stable and the SPF database is updated.
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Roll-back Procedures

- Step: Revert BGP session, Command: show bgp *, Description: Revert the BGP session to its previous state if the resolution step fails. This command will display the BGP session information and allow for manual intervention.

- Step: Revert OSPF adjacency, Command: show ospf adjacency *, Description: Revert the OSPF adjacency to its previous state if the resolution step fails. This command will display the OSPF adjacency information and allow for manual intervention.

- Step: Revert ISIS LSP database, Command: show isis database, Description: Revert the ISIS LSP database to its previous state if the resolution step fails. This command will display the ISIS LSP database information and allow for manual intervention.